

SCHOOL AHI
GCE O-LEVEL PRELIMINARY EXAMINATION 2023
SECONDARY 4 EXPRESS

Name:	Class:	Register No.:
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ADDITIONAL MATHEMATICS
Paper 2

4049/02
11 August 2023

Candidates answer on the Question Paper.

2 hours 15 minutes

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions.

Give non-exact numerical answers to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
The use of an approved scientific calculator is expected, where appropriate.
You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.
The total number of marks for this paper is 90.

For Examiner's Use
/90

Mathematical Formulae

1. ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{r}a^{n-r}b^r + \dots + b^n$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)\dots(n-r+1)}{r!}$

2. TRIGONOMETRY

Identities

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formulae for $\triangle ABC$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2}bc \sin A$$

- 1 (a) Find the range of values of x for which the expression $3 - 2x^2$ is negative. [2]

- (b) Find the set of values of the constant k for which the curve $y = x^2$ lies entirely above the line $y = k(x + 1)$. [3]

- 2 (a)** Find the range of values of k such that the line $x + y = 3$ intersects the curve $x^2 - 2x + 2y^2 = k$. [4]

- (b)** State a possible value of k if there is no intersection between the line and the curve. [1]

3 A polynomial, P , is $x^{2n} - (k+1)x^2 + k$ where n and k are positive integers.

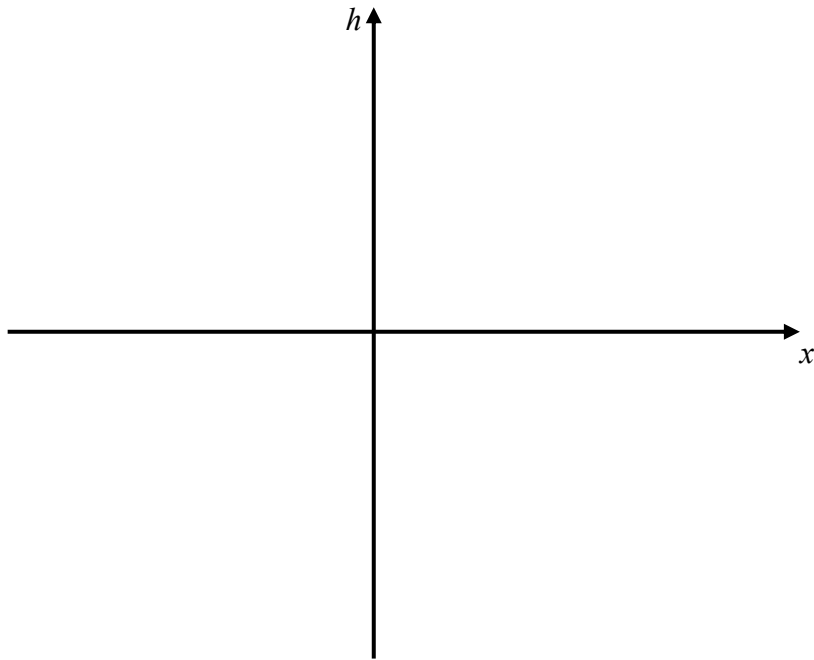
(a) Explain why $x-1$ is a factor of P for all values of k . [2]

(b) Given that $k = 4$, find the value of n for which $x-2$ is a factor of P .
Hence factorise P completely. [4]

- 4 A projectile was launched from a catapult to hit a defence structure on a fort. The height, h metres, of the projectile above ground is given by the equation $h = -2x^2 + 3x + 1.5$, where x metres is the horizontal distance from the catapult.
- (i) By expressing the function in the form $h = a(x - m)^2 + n$, where a , m and n are constants, explain whether the projectile can reach a height of 3 metres. [2]
- (ii) Given that the defence structure is 1.4 metres horizontally from the catapult and 0.8 metres above the ground, justify if the projectile will hit the structure. [2]

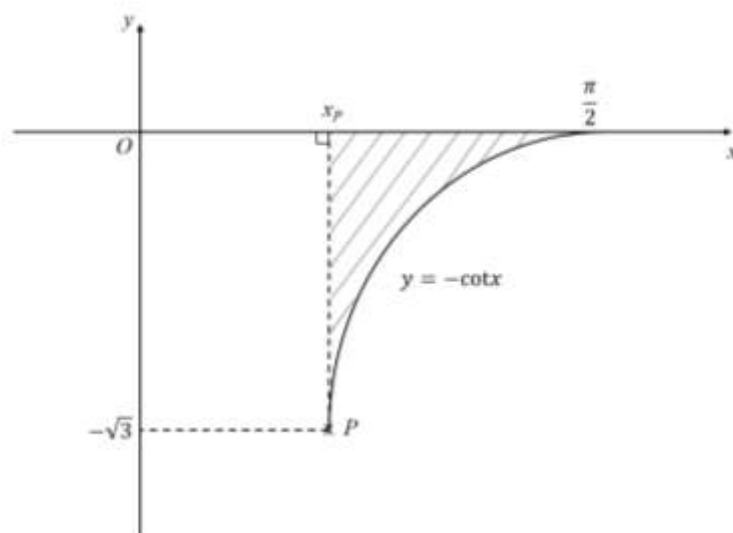
(iii) Sketch the curve of $h = -2t^2 + 3t + 1.5$.

[2]



- 5 (a) Differentiate $\ln(\sin x)$ with respect to x . [2]

(b)



The diagram shows part of the curve $y = -\cot x$, cutting the x -axis at $\left(\frac{\pi}{2}, 0\right)$.

The line $y = -\sqrt{3}$ intersects the curve at P .

- (i) State the value of x_p , the x -coordinate of P . [1]

- (ii) Explain why the expression $\int_{x_p}^{\frac{\pi}{2}} -\cot x \, dx$ does not give the area of the shaded region. [1]

(iii) Find the exact area of the shaded region.

[3]

5 (a) Without using a calculator, show that $\cos\left(\frac{7\pi}{12}\right) = \frac{1}{4}(\sqrt{2} - \sqrt{6})$. [3]

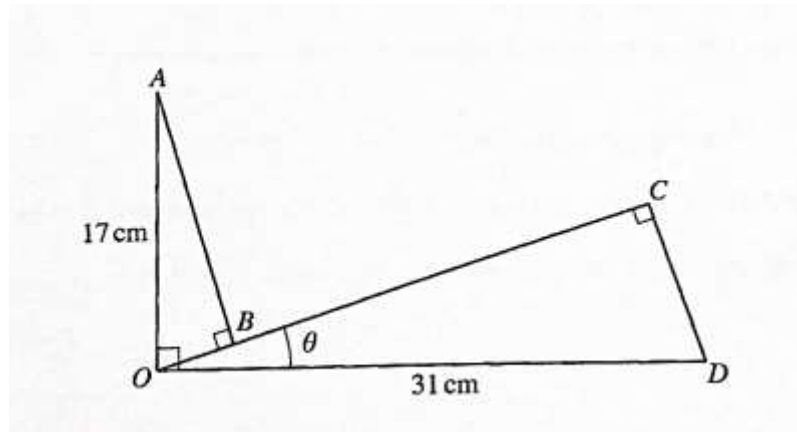
(b) Evaluate $\int_0^{\frac{\pi}{12}} 3\cos^2 x - \sin^2 x \, dx$ exactly. [4]

- 6 (a) (i) Factorise $x^6 - 64$ completely. [2]

(ii) Hence solve $x^6 - 64 = (x^2 + 4)^2 - (2x)^2$. [3]

- (b) Find the values of the integers a and b for which $\frac{a + \sqrt{b}}{2}$ is the solution of the equation $2x\sqrt{3} + x\sqrt{125} = x\sqrt{45} + \sqrt{12}$. [4]

7



The diagram shows three fixed points O , A and D such that $OA = 17$ cm, $OD = 31$ cm and angle $AOD = 90^\circ$.

The lines AB and DC are perpendicular to the line OC which makes an angle θ with the line OD .

The angle θ can vary in such a way that the point B lies between the points O and C .

- (i) Show that $AB + BC + CD = (48 \cos \theta + 14 \sin \theta)$ cm. [3]

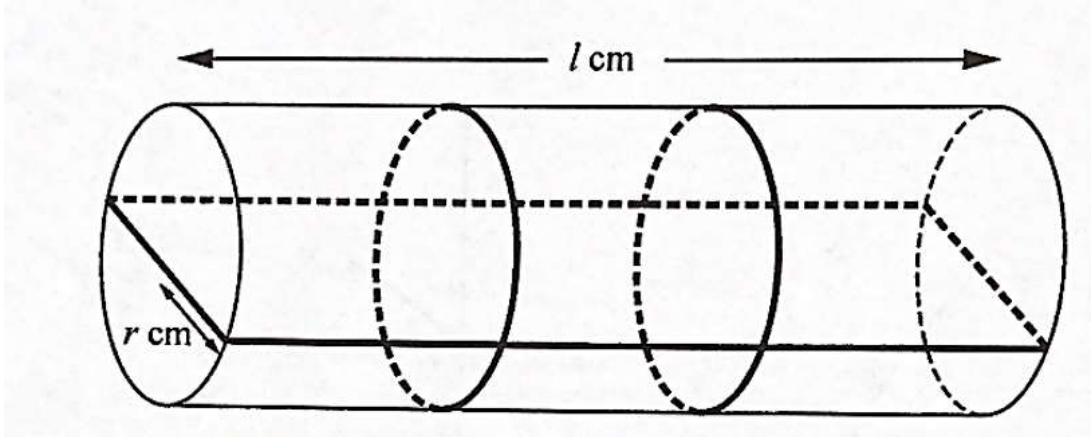
(ii) Find the values of θ for which $AB + BC + CD = 49$ cm.

[6]

(iii) Find the maximum value of $AB + BC + CD$ and the corresponding value of θ .

[2]

- 8 The diagram shows a roll of material in the shape of a cylinder of radius r cm and length l cm.
 The roll is held together by three pieces of adhesive tape whose width and thickness may be ignored.
 One piece of tape is in the shape of a rectangle, the other two pieces are in the shape of circles.
 The total length of tape is 600 cm.



- (i) Show that the volume, $V \text{ cm}^3$, of the cylinder is given by
 $V = \pi r^2 (300 - 2r - 2\pi r)$.

[3]

- (ii) Given that r can vary, show that V has a stationary value when $r = \frac{k}{1+\pi}$, where k is a constant to be found, and find the corresponding value of l . [5]

- (iii) Determine if the volume is a minimum or maximum. [3]

- 9 A particle travelling in a straight line passes through a fixed point O with a speed of 8 m/s.
The acceleration, a m/s², of the particle t s after passing through O , is given by $a = -e^{-0.1t}$.

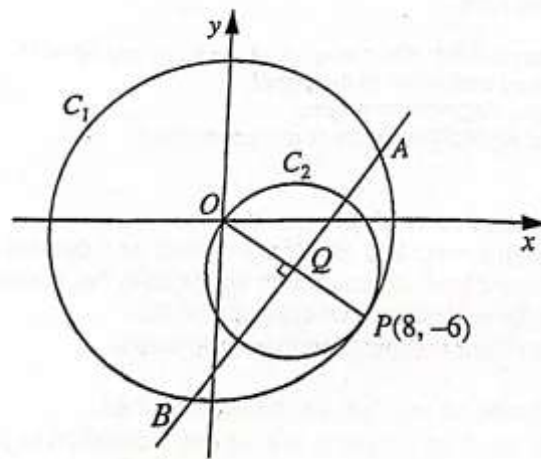
The particle comes to instantaneous rest at the point P .

- (i) Show that the particle reaches P when $t = 10 \ln 5$. [5]

- (ii) Calculate the distance OP . [3]

- (iii) Explain why the particle is again at O at some instant during the fiftieth second after first passing through O . [3]

10



The diagram shows two circles C_1 and C_2 .

Circle C_1 has its centre at the origin O .

Circle C_2 passes through O and has its centre at Q .

The point $P(8, -6)$ lies on both circles and OP is a diameter of C_2 .

(a) Find the equation of C_1 .

[2]

(b) Explain why the equation of C_2 is $x^2 + y^2 - 8x + 6y = 0$.

[3]

- (c) The line through Q perpendicular to OP meets the circle C_1 at the point A and B .
Show that the x -coordinates of A and B are $a+b\sqrt{3}$ and $a-b\sqrt{3}$ respectively,
where a and b are integers to be found. [7]

END OF PAPER

[Turn over